

5.2 INFLIGHT POWER CHECKS

The power assurance check is provided to ensure that the engines are capable of developing the power used in establishing the performance data presented in this manual.

The engine trend check is provided to periodically monitor the long-term engine behavior.

5.2.1 Power Assurance Check

This check is performed in cruise flight with one engine operating at IDLE; each engine being checked separately, in turn. The generator and non-essential bleed air consumers (if installed) of the check engine must be switched off during its check. Engine anti-icing shall be on if ambient conditions require its use.

The power assurance check must be performed at intervals not to exceed 100 flight hours or whenever abnormal engine function is noted.

1. Note altimeter pressure setting; set for pressure altitude (29.92 in. Hg or 1013 hPa).

WARNING MAINTAIN A SUITABLE MINIMUM SAFE HEIGHT IN CASE OF ENGINE FAILURE.

2. Select a convenient pressure altitude in increments of 1000 ft from sea level up to the maximum check altitude of 10,000 ft. Establish and maintain level flight with 60 KIAS (V_Y).

WARNING SYSTEMS OR COMPONENTS REQUIRED FOR SAFE FLIGHT SHALL REMAIN ON.

3. Switch off all non-essential bleed air consumers. If engine anti-icing is frequently required for safe flight, it should be operated during each power check.
4. Reduce electrical load as much as possible.
5. Switch off generator of the engine to be checked.
6. Set power lever of non-check engine slowly to IDLE.

NOTE Observe limits (N_1 , TOT, Torque) of check engine.

7. Set N_2/N_{RO} to 100%.

WARNING DO NOT EXCEED ENGINE OR HELICOPTER LIMITS.

8. Adjust collective lever as necessary to achieve an indicated torque of 95% **or** an indicated TOT of 810 °C, **whichever is reached first**. Adjust airspeed as necessary to maintain level flight.
9. Allow check engine to stabilize at the adjusted power setting for at least 1 minute.

10. Record the following values (avoid parallax errors):
 - Pressure Altitude (FT)
 - Outside Air Temp (°C)
 - Torque (%)
 - TOT (°C)
 - N1 (%)
 - Engine Anti-icing (on or off)
 11. Set power lever of non-check engine to FLIGHT.
 12. Switch on generator of checked engine.
 13. Repeat above steps 5. through 12. to check other engine.
 14. Reset altimeter and, if necessary, switch on bleed air consumers.
 15. After flight, use the appropriate Inflight Power Check diagram (see Fig. 5-1 or 5-1A) to determine either maximum TOT or minimum required torque available under the prevailing ambient conditions.
 16. Power is assured and all performance data contained in this manual can be achieved when each engine's:
 - indicated TOT is equal to or less than the chart derived maximum TOT value, or
 - indicated torque is equal to or greater than the chart derived minimum required torque value.
- NOTE** If power is not assured on one or both engines then the total power available is less than the minimum specification and all performance data contained in this manual cannot be achieved. Refer to appropriate maintenance manual to determine cause of power loss.

EXAMPLE: (see Fig. 5-1)

Determine: **Engine power assurance - Torque limit reached first**

Known:	Indicated torque	95 %
	Pressure altitude	4000 ft
	OAT	-30 °C
	Indicated TOT	730 °C
	Anti-icing	OFF

- Solution:
1. Enter chart at known torque (95%).
 2. Move vertically up to known pressure altitude (4000 ft) and then trace horizontally left through the TOT field.
 3. Reenter chart at known OAT (-30 °C) and move vertically up to intersect the above tracing.
 4. At point of intersection read maximum TOT = 740 °C.

Since the indicated TOT (730 °C) is lower than the obtained chart TOT limit (740 °C), the power assurance check for this engine is acceptable.

EXAMPLE: (see Fig. 5-1)

Determine: **Engine power assurance - TOT limit reached first**

Known:	Indicated TOT	810 °C
	Pressure altitude	4000 ft
	OAT	20 °C
	Indicated torque	90 %
	Anti-icing	OFF

- Solution:
1. Enter chart at known OAT (20 °C).
 2. Move vertically up to known indicated TOT (810 °C) and trace horizontally right to intercept known PA (4000 ft).
 3. From this intercept point move vertically down to read minimum required torque = 86 %.

Since the indicated torque (90 %) is higher than the obtained chart minimum required torque (86 %), the power assurance check for this engine is acceptable.

INFLIGHT POWER CHECK

ALLISON 250-C20B

N_2 – 100%

N_1 – 95% OR ABOVE

GENERATOR – OFF

ANTI-ICING – OFF

BLEED AIR CONSUMERS – OFF

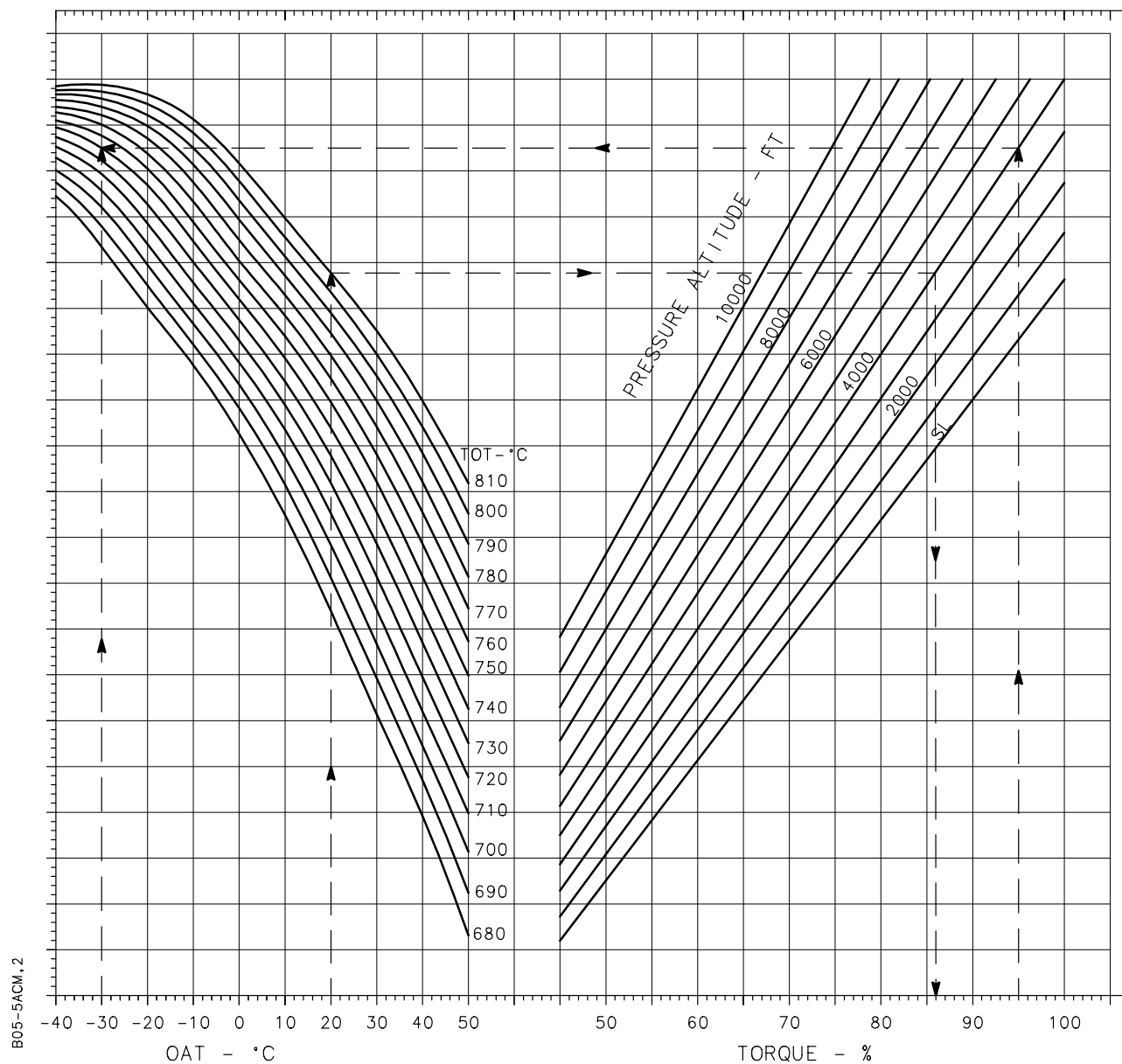


Fig. 5-1 Inflight Power Check (Anti-icing - OFF)

LBA APPROVED

Rev. 3

INFLIGHT POWER CHECK

ALLISON 250-C20B

N_2 – 100%

N_1 – 95% OR ABOVE

GENERATOR – OFF

ANTI-ICING – ON

BLEED AIR CONSUMERS – OFF

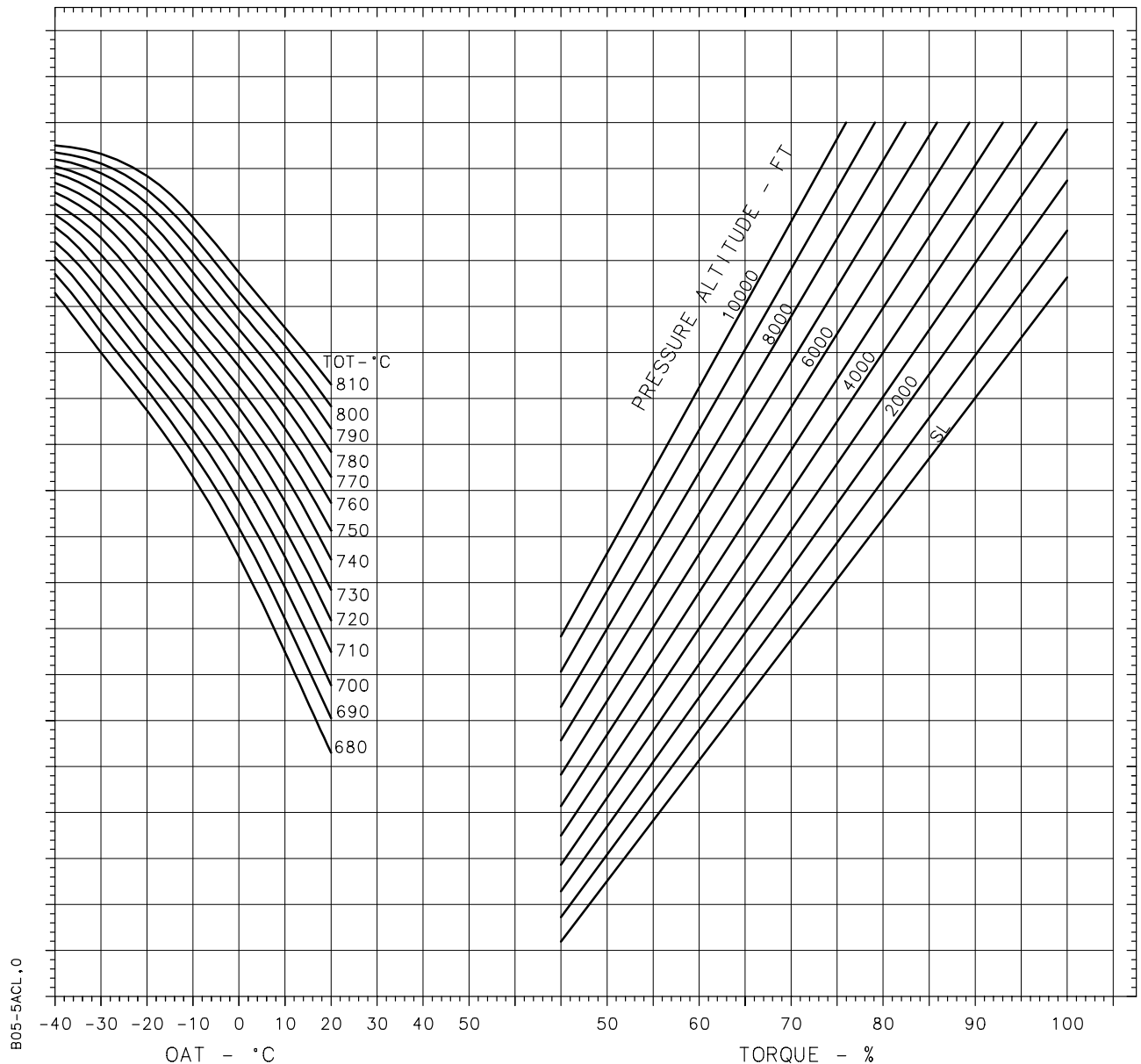


Fig. 5-1A Inflight Power Check (Anti-icing - ON)

5.2.2 Engine Trend Check

NOTE The engine trend check is not an alternate procedure to the power assurance check. Only the results of the Power Assurance Check (see Para 5.2.1) shall be used to ensure that the engines are capable of developing the power necessary to meet the presented performance data.

The engine trend check is provided for the purpose of monitoring the long-term behaviour and recognizing sudden significant trend-line deviations of the engine performance at minimum operator expense. This check, which should be repeated on a regular basis based upon operator experience, is performed in cruise flight with both engines operating; each engine being checked separately, in turn. The non-essential bleed air consumers (if installed) of the check engine must be switched off during its check. Engine anti-icing shall be on if ambient conditions require its use.

1. Note altimeter pressure setting; set for pressure altitude (29.92 in. Hg or 1013 hPa).
2. Select a convenient pressure altitude in increments of 1000 ft from sea level up to the maximum check altitude of 10,000 ft. Establish and maintain level flight with 60 KIAS (V_Y) or more.

WARNING SYSTEMS OR COMPONENTS REQUIRED FOR SAFE FLIGHT SHALL REMAIN ON.

3. Switch off all non-essential bleed air consumers. If engine anti-icing is frequently required for safe flight, it should be operated during each power check.
4. Reduce electrical consumption as much as possible.
5. Set N_2 at 100%.
6. Adjust collective lever as necessary to achieve an indicated torque of 86% **or** an indicated TOT of 779 °C, **whichever is reached first**.

NOTE If N_1 is not at least 95%, the engine bleed valve is open and the check result is not valid. In this case the check altitude should be increased.

7. Allow check engine to stabilize at the adjusted power setting for at least 1 minute.
8. Record the following values (avoid parallax errors):
 - Pressure Altitude (FT)
 - Outside Air Temp (°C)
 - Torque (%)
 - TOT (°C)
 - N_1 (%)
 - Engine Anti-icing (on or off)
9. Repeat above steps 4. through 8. to check other engine.

10. Reset altimeter and, if necessary, switch on bleed air consumers.
11. Following flight, use the appropriate Inflight Power Check diagram (see Fig. 5-1 or 5-1A) to determine the minimum required torque available under the prevailing ambient conditions.

5.2.3 Power Trend Monitoring

A power trend monitoring shall be established to observe the deterioration of the engines with time in service as well as potential engine malfunctions.

NOTE The power trend check preconditions and the power trend check procedure have to be followed very accurately, otherwise the trend monitoring will not be reliable.

To establish a power trend chart, the values of Δ Torque (defined as measured torque minus chart derived minimum required torque) have to be recorded versus time in service (see Fig. 5-2).

CAUTION MAINTENANCE ACTION MUST BE CONSIDERED IF A Δ TORQUE VALUE DEVIATES SIGNIFICANTLY FROM THE ESTABLISHED TREND LINE.

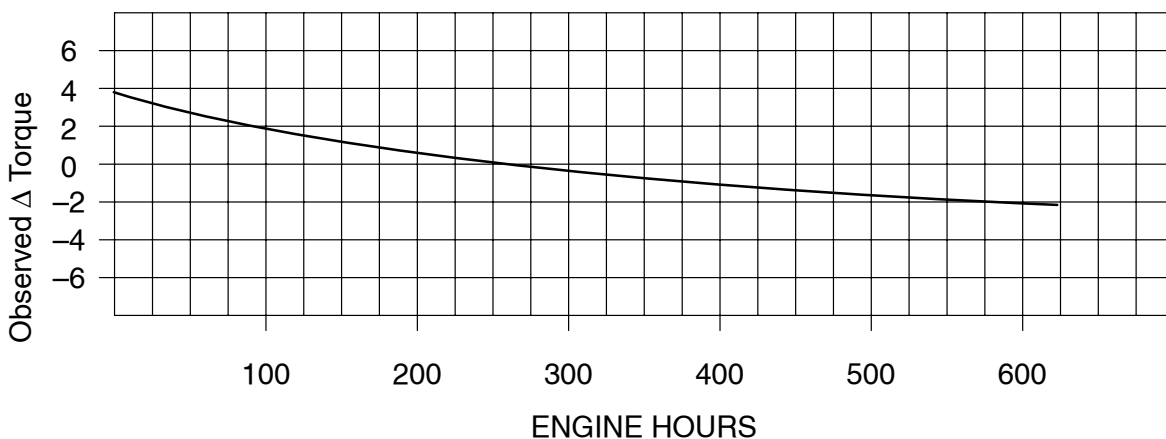


Fig. 5-2 Plotting Example for Power Trend Monitoring